

Simplified Luminosity Function

An explicit derivation and exploration of the detected population

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Abstract

We replace the bounded luminosity distribution by a power-law event-rate density extending over all positive luminosities. The purpose is to understand which luminosities contribute to a survey and how the detection rate changes with limiting flux, cadence, and the slope of the luminosity function. We begin with the detectable volume of an event of luminosity L , derive the luminosity and distance integrals in both orders, and then evaluate the viewing-angle integral explicitly. This separates an exact result from the project's most-contributing-point approximation. The same reasoning gives the detected luminosity, distance, and angle distributions, and explains precisely when the threshold at D_{Euc} is a useful characteristic luminosity. The final steps connect the result to repeated detections and to the original bounded luminosity function. Only a few numerical examples are retained to illustrate the analytic behavior.

1 What is being simplified?

The previous derivation [2] distributes the on-axis spectral luminosity at the deceleration time,

$$L \equiv L_\nu(t_{\text{dec}}), \quad [L] = \text{erg s}^{-1} \text{ Hz}^{-1}. \quad (1)$$

It specifies a total intrinsic rate and a normalized luminosity distribution between L_{min} and L_{max} . Here we specify directly how many events occur in each luminosity interval, per unit volume and time:

$$\boxed{\frac{d\mathcal{R}}{dL} = \frac{\mathcal{A}}{L_{\text{ref}}} \left(\frac{L}{L_{\text{ref}}} \right)^\alpha, \quad 0 < L < \infty, \quad L_{\text{ref}} = 10^{32} \text{ erg s}^{-1} \text{ Hz}^{-1}.} \quad (2)$$

The two adjustable population parameters are $\mathcal{A}$ and α . The reference luminosity L_{ref} is fixed; it is a convention for quoting $\mathcal{A}$, not another physical luminosity scale. Multiplying by L gives

$$\frac{d\mathcal{R}}{d \ln L} = \mathcal{A} \left(\frac{L}{L_{\text{ref}}} \right)^{\alpha+1}, \quad \left. \frac{d\mathcal{R}}{d \ln L} \right|_{L=L_{\text{ref}}} = \mathcal{A}. \quad (3)$$

Thus $\mathcal{A}$ is the volumetric event rate per natural-log luminosity interval at L_{ref} . It can be quoted in $\text{Gpc}^{-3} \text{ yr}^{-1}$. The power-law index is defined *per unit luminosity*: per logarithmic interval the index is $\alpha + 1$. We keep this distinction throughout.

We use the notation of the thesis and the previous LF derivation. In particular D_{Euc} is the Euclidean radius, $q = \theta_{\text{obs}}/\theta_j$ is the viewing angle in units of the jet half-angle, F_{lim} is the spectral limiting flux, and i is the required number of detections. The physical luminosity thresholds L_{dec} , L_j , L_{nr} , L_i , and $L_{\text{Euc}}(q)$ are defined as we encounter them. No auxiliary angular integral or abbreviated exponent is needed. All rates below are all-sky rates; a uniform surveyed solid angle multiplies them by its fraction of 4π .

The equations use consistent length units. For example, if luminosities and fluxes are in cgs units, use distances in cm and convert the quoted $\mathcal{A}$ from $\text{Gpc}^{-3} \text{ yr}^{-1}$ to $\text{cm}^{-3} \text{ yr}^{-1}$ before multiplying by a volume in cm^3 . This conversion is understood whenever $4\pi D^2 F_{\text{lim}}$ and $D^3 \mathcal{A}$ appear together.

1.1 What remains the same as in the project

The source population is uniform in Euclidean volume, with random jet orientations and a common afterglow light-curve shape. Changing L multiplies the entire light curve without changing p , Γ_0 , θ_j , t_{dec} , or t_j . Here p is the electron-energy power-law index of the project; it is not a probability-density symbol. The model is therefore a brightness distribution at fixed dynamics. Correlations between luminosity and duration, jet opening angle, or spectral evolution would require a further extension.

1.2 Why there is no finite total intrinsic rate

At low luminosity, $\int_0 L^\alpha dL$ is finite only for $\alpha > -1$. At high luminosity, $\int^\infty L^\alpha dL$ is finite only for $\alpha < -1$. No slope satisfies both conditions. Consequently Eq. (2) is an event-rate density, not a normalized probability distribution with a finite R_{int} . Its total luminosity output is also divergent: $\int_0^\infty L^{\alpha+1} dL$ has incompatible conditions at its two ends.

The question is instead whether the *detected* rate is finite. Very faint sources occupy a very small detectable volume. That extra volume factor is what can make the observation integral converge, despite the divergent intrinsic population. We now derive it rather than assume it.

2 From the light curve to a physical luminosity threshold

The thesis [1] approximates an off-axis light curve as joining the on-axis light curve at the peak time $t_p(q)$. The relevant angles and times are

$$q_{\text{dec}} = 1 + \frac{1}{\Gamma_0 \theta_j}, \quad q_j = 2, \quad q_{\text{nr}} = \frac{\sqrt{2}}{\theta_j}, \quad (4)$$

$$t_j = t_{\text{dec}} (\Gamma_0 \theta_j)^{8/3}, \quad t_{\text{nr}} = t_j (q_{\text{nr}} - 1)^2, \quad (5)$$

$$t_p(q) = \begin{cases} t_{\text{dec}}, & 0 \leq q \leq q_{\text{dec}}, \\ t_j (q - 1)^{8/3}, & q_{\text{dec}} < q < 2, \\ t_j (q - 1)^2, & 2 \leq q \leq q_{\text{nr}}. \end{cases} \quad (6)$$

For the optical spectral segment used in the project, the on-axis decline, normalized to the deceleration flux, is

$$\tilde{F}_\nu(t) = \begin{cases} (t/t_{\text{dec}})^{-3(p-1)/4}, & t_{\text{dec}} \leq t < t_j, \\ (t_j/t_{\text{dec}})^{-3(p-1)/4} (t/t_j)^{-p}, & t_j \leq t < t_{\text{nr}}, \\ (t_j/t_{\text{dec}})^{-3(p-1)/4} (t_{\text{nr}}/t_j)^{-p} (t/t_{\text{nr}})^{-(15p-21)/10}, & t \geq t_{\text{nr}}. \end{cases} \quad (7)$$

The last line is the thesis's Newtonian decline. It is needed when repeated visits occur after t_{nr} ; peak detection alone uses only $t_p \leq t_{\text{nr}}$. For a threshold time earlier than t_{dec} , the peak cap is $\tilde{F}_\nu = 1$. There is no resolved pre-deceleration rise in this approximation.

Substituting $t_p(q)$ gives the normalized peak flux:

$$\tilde{F}_p(q) = \tilde{F}_\nu(t_p(q)) = \begin{cases} 1, & 0 \leq q \leq q_{\text{dec}}, \\ (\Gamma_0 \theta_j)^{-2(p-1)} (q - 1)^{-2(p-1)}, & q_{\text{dec}} < q < 2, \\ (\Gamma_0 \theta_j)^{-2(p-1)} (q - 1)^{-2p}, & 2 \leq q \leq q_{\text{nr}}. \end{cases} \quad (8)$$

An event of luminosity L at distance D therefore has

$$F_p(q, D; L) = \frac{L \tilde{F}_p(q)}{4\pi D^2}. \quad (9)$$

Solving $F_p = F_{\text{lim}}$ for its maximal detectable distance gives

$$D_{\text{max}}(q; L) = \left(\frac{L \tilde{F}_p(q)}{4\pi F_{\text{lim}}} \right)^{1/2}. \quad (10)$$

Now ask the converse question: *at a given viewing angle, what luminosity is just detectable at D_{Euc} ?* This is the threshold already called $L_{\text{Euc}}(q)$ in the previous LF derivation:

$$L_{\text{Euc}}(q) = \frac{4\pi D_{\text{Euc}}^2 F_{\text{lim}}}{\tilde{F}_p(q)} = \frac{L_{\text{dec}}}{\tilde{F}_p(q)}, \quad L_{\text{dec}} = 4\pi D_{\text{Euc}}^2 F_{\text{lim}}. \quad (11)$$

In the same physical notation,

$$L_j = L_{\text{dec}}(\Gamma_0 \theta_j)^{2(p-1)}, \quad L_{\text{nr}} = L_j(q_{\text{nr}} - 1)^{2p}, \quad (12)$$

$$L_{\text{Euc}}(q) = \begin{cases} L_{\text{dec}}, & 0 \leq q \leq q_{\text{dec}}, \\ L_j(q - 1)^{2(p-1)}, & q_{\text{dec}} < q < 2, \\ L_j(q - 1)^{2p}, & 2 \leq q \leq q_{\text{nr}}. \end{cases} \quad (13)$$

Each threshold has a direct meaning: L_{dec} reaches the boundary within the constant-peak-flux core, L_j reaches it at $q = 2$, and L_{nr} reaches it at the outer angular boundary. These are detection thresholds, not lower or upper bounds on the intrinsic luminosity function.

The full-time knee at the right of thesis Figure 15 corresponds to $\tilde{F}_{\text{lim,dec}} = 1$, or $L = L_{\text{dec}}$. The associated angle is $q_{\text{dec}} = 1 + (\Gamma_0 \theta_j)^{-1} \simeq 1$. Events with $L < L_{\text{dec}}$ are still detectable at smaller distances. Thus L_{dec} is the lowest luminosity visible *at the Euclidean boundary*, not the lowest luminosity in the detected sample.

3 The detection-rate integral, step by step

The thesis orientation measure is $\theta_j^2 q dq$, with $\theta_j^2 \int_0^{q_{\text{nr}}} q dq = 1$. The radial volume element is $4\pi D^2 dD$. Multiplying the orientation, volume, and luminosity-rate elements gives

$$R_{\text{det}} = 4\pi \theta_j^2 \int_0^{q_{\text{nr}}} q dq \int_0^{D_{\text{Euc}}} D^2 dD \int_0^\infty \frac{\mathcal{A}}{L_{\text{ref}}} \left(\frac{L}{L_{\text{ref}}} \right)^\alpha \Theta \left(\frac{L \tilde{F}_p(q)}{4\pi D^2} - F_{\text{lim}} \right) dL. \quad (14)$$

The step function includes an event precisely when its peak exceeds the limiting flux. Every factor in this expression comes directly from the population or the survey geometry. We evaluate it in two orders to expose the physical origin of the result.

3.1 Distance first: the detectable volume

For a fixed q and L , the distance integral ends at the smaller of $D_{\text{max}}(q; L)$ and D_{Euc} . Hence

$$\begin{aligned} \int_0^{D_{\text{Euc}}} 4\pi D^2 \Theta(D_{\text{max}} - D) dD &= \frac{4\pi}{3} \min[D_{\text{max}}(q; L)^3, D_{\text{Euc}}^3] \\ &= \frac{4\pi D_{\text{Euc}}^3}{3} \min \left[\left(\frac{L}{L_{\text{Euc}}(q)} \right)^{3/2}, 1 \right]. \end{aligned} \quad (15)$$

Below $L_{\text{Euc}}(q)$, increasing the luminosity increases the horizon as $L^{1/2}$ and the detectable volume as $L^{3/2}$. Above $L_{\text{Euc}}(q)$, the entire Euclidean volume at that angle is already visible, so there is no further volume gain.

Splitting the luminosity integral at this physical threshold gives

$$R_{\text{det}} = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{3} \int_0^{q_{\text{nr}}} q dq \left[\int_0^{L_{\text{Euc}}(q)} \frac{dL}{L_{\text{ref}}} \left(\frac{L}{L_{\text{ref}}} \right)^\alpha \left(\frac{L}{L_{\text{Euc}}(q)} \right)^{3/2} + \int_{L_{\text{Euc}}(q)}^\infty \frac{dL}{L_{\text{ref}}} \left(\frac{L}{L_{\text{ref}}} \right)^\alpha \right]. \quad (16)$$

The first integral equals

$$\left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} \frac{1}{\alpha + 5/2}, \quad \alpha > -5/2, \quad (17)$$

and the second equals

$$\left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} \frac{1}{-\alpha - 1}, \quad \alpha < -1. \quad (18)$$

Both are finite precisely when

$$\boxed{-\frac{5}{2} < \alpha < -1.} \quad (19)$$

This is the key difference between the intrinsic and detected populations. The faint intrinsic density is multiplied by $L^{3/2}$ before it is integrated; the bright intrinsic density is multiplied by a volume that has saturated.

Adding the two contributions yields

$$\begin{aligned} R_{\text{det}} &= \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{3} \left(\frac{1}{\alpha + 5/2} + \frac{1}{-\alpha - 1} \right) \int_0^{q_{\text{nr}}} q \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} dq \\ &= \boxed{\frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{(-\alpha - 1)(2\alpha + 5)} \int_0^{q_{\text{nr}}} q \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} dq.} \end{aligned} \quad (20)$$

Only the viewing-angle integration remains. No most-contributing-point approximation has been made.

3.2 Luminosity first: why the distance dependence is simple

At a fixed distance and angle the lowest detectable luminosity is $L_{\text{Euc}}(q)(D/D_{\text{Euc}})^2$. Thus

$$\begin{aligned} &\int_{L_{\text{Euc}}(q)(D/D_{\text{Euc}})^2}^\infty \frac{\mathcal{A}}{L_{\text{ref}}} \left(\frac{L}{L_{\text{ref}}} \right)^\alpha dL \\ &= \frac{\mathcal{A}}{-\alpha - 1} \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} \left(\frac{D}{D_{\text{Euc}}} \right)^{2\alpha+2}. \end{aligned} \quad (21)$$

After multiplying by the shell volume, the radial integrand is proportional to $D^{2\alpha+4}$. Its integral is

$$\int_0^{D_{\text{Euc}}} D^2 \left(\frac{D}{D_{\text{Euc}}} \right)^{2\alpha+2} dD = \frac{D_{\text{Euc}}^3}{2\alpha + 5}, \quad \alpha > -5/2. \quad (22)$$

Inserting this into Eq. (14) reproduces Eq. (20). The faint-luminosity divergence has appeared as a small-distance divergence: extremely faint sources can contribute only in an extremely small nearby volume.

3.3 Exact dependence on normalization, depth, and Euclidean radius

Every $L_{\text{Euc}}(q)$ is proportional to $D_{\text{Euc}}^2 F_{\text{lim}}$. At fixed light-curve shape and selection, Eq. (20) therefore gives

$$\boxed{R_{\text{det}} \propto \mathcal{A} F_{\text{lim}}^{\alpha+1} D_{\text{Euc}}^{2\alpha+5}.} \quad (23)$$

For example, $\alpha = -2$ gives $R_{\text{det}} \propto \mathcal{A} D_{\text{Euc}}/F_{\text{lim}}$. The radius exponent combines the larger source volume with the greater luminosity required to reach the new boundary. It is not simply the volume exponent three.

A change of L_{ref} must not change the physical intensity. If the same population is quoted at a different reference luminosity,

$$\mathcal{A}' = \mathcal{A} \left(\frac{L'_{\text{ref}}}{L_{\text{ref}}} \right)^{\alpha+1}. \quad (24)$$

The combination $\mathcal{A}(L_{\text{Euc}}/L_{\text{ref}})^{\alpha+1}$ then stays unchanged. In particular, setting the reference equal to a survey-dependent threshold while keeping its amplitude fixed would change the population.

4 Which luminosities actually contribute?

4.1 Contributions per logarithmic luminosity interval

A logarithmic interval is the natural way to compare decades of luminosity. Multiplying the differential rate by L gives

$$\frac{d^2 R_{\text{det}}}{dq d \ln L} = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{3} q \left(\frac{L}{L_{\text{ref}}} \right)^{\alpha+1} \begin{cases} (L/L_{\text{Euc}}(q))^{3/2}, & L < L_{\text{Euc}}(q), \\ 1, & L \geq L_{\text{Euc}}(q). \end{cases} \quad (25)$$

At any fixed angle, the logarithmic slope is consequently

$$\frac{d \ln[d^2 R_{\text{det}}/(dq d \ln L)]}{d \ln L} = \begin{cases} \alpha + 5/2 > 0, & L < L_{\text{Euc}}(q), \\ \alpha + 1 < 0, & L > L_{\text{Euc}}(q). \end{cases} \quad (26)$$

The contribution rises up to $L_{\text{Euc}}(q)$ and falls beyond it. The luminosity needed to reach the Euclidean boundary is therefore the *maximum of the contribution per logarithmic luminosity interval at that angle*. Within $q \leq q_{\text{dec}}$, that maximum is at $L = L_{\text{dec}}$.

This establishes the proposed threshold argument, but not a narrow concentration. A slowly declining tail can contain most events even when its density nowhere exceeds the value at the threshold. We can quantify that distinction without choosing any numerical afterglow parameters.

4.2 Fractions and medians, derived by integration

Dividing Eq. (25) by dR_{det}/dq gives the normalized conditional logarithmic distribution:

$$\frac{d^2 R_{\text{det}}/(dq d \ln L)}{dR_{\text{det}}/dq} = \frac{2(-\alpha - 1)(\alpha + 5/2)}{3} \begin{cases} (L/L_{\text{Euc}}(q))^{\alpha+5/2}, & L \leq L_{\text{Euc}}(q), \\ (L/L_{\text{Euc}}(q))^{\alpha+1}, & L \geq L_{\text{Euc}}(q). \end{cases} \quad (27)$$

Here *conditional* means that the viewing angle is held fixed. Integrating this density from zero luminosity to L gives

$$\Pr(L' < L \mid q, \text{detected}) = \begin{cases} \frac{2(-\alpha - 1)}{3} (L/L_{\text{Euc}}(q))^{\alpha+5/2}, & L \leq L_{\text{Euc}}(q), \\ 1 - \frac{2(\alpha + 5/2)}{3} (L/L_{\text{Euc}}(q))^{\alpha+1}, & L \geq L_{\text{Euc}}(q). \end{cases} \quad (28)$$

The two branches agree at the threshold. In particular,

$$\Pr(L < L_{\text{Euc}}(q) \mid q, \text{detected}) = \frac{2(-\alpha - 1)}{3}, \quad (29)$$

$$\Pr(L > L_{\text{Euc}}(q) \mid q, \text{detected}) = \frac{2(\alpha + 5/2)}{3}. \quad (30)$$

Their sum is one. The lower-luminosity side contains more than half the events for $\alpha < -7/4$; the brighter side contains more than half for $\alpha > -7/4$. At $\alpha = -7/4$ the threshold is also the median.

Setting Eq. (28) equal to $1/2$ yields

$$\frac{L_{\text{med}|q}}{L_{\text{Euc}}(q)} = \begin{cases} \left[\frac{3}{4(-\alpha - 1)} \right]^{1/(\alpha+5/2)}, & -5/2 < \alpha \leq -7/4, \\ \left[\frac{4(\alpha + 5/2)}{3} \right]^{-1/(\alpha+1)}, & -7/4 \leq \alpha < -1. \end{cases} \quad (31)$$

For $\alpha = -2$, the median is $9L_{\text{Euc}}(q)/16$: the most contributing logarithmic interval and the median need not coincide. For any physical limits $L_{\text{lower}} < L_{\text{Euc}}(q) < L_{\text{upper}}$, the enclosed fraction follows directly:

$$\begin{aligned} & \Pr(L_{\text{lower}} < L < L_{\text{upper}} \mid q, \text{detected}) \\ &= 1 - \frac{2(-\alpha - 1)}{3} \left(\frac{L_{\text{lower}}}{L_{\text{Euc}}(q)} \right)^{\alpha+5/2} - \frac{2(\alpha + 5/2)}{3} \left(\frac{L_{\text{upper}}}{L_{\text{Euc}}(q)} \right)^{\alpha+1}. \end{aligned} \quad (32)$$

At $\alpha = -2$, the interval from $0.1L_{\text{Euc}}(q)$ to $10L_{\text{Euc}}(q)$ contains about 75.6% of detections at that angle. Even this central example spans two decades. As either convergence boundary is approached, one of the powers in Eq. (32) tends to zero and the corresponding tail becomes increasingly broad.

4.3 Combining viewing angles and the existence of moments

Each angle has a different threshold and a different total detected weight. The global luminosity CDF must therefore be the weighted mixture

$$\Pr(L' < L \mid \text{detected}) = \frac{\int_0^{q_{\text{nr}}} q (L_{\text{Euc}}(q)/L_{\text{ref}})^{\alpha+1} \Pr(L' < L \mid q, \text{detected}) dq}{\int_0^{q_{\text{nr}}} q (L_{\text{Euc}}(q)/L_{\text{ref}})^{\alpha+1} dq}. \quad (33)$$

The global median is the luminosity at which this CDF equals $1/2$. It is not obtained by averaging conditional medians. Nor must the maximum of the global contribution per logarithmic luminosity interval be at L_{dec} : more off-axis events have larger $L_{\text{Euc}}(q)$. Equation (33) is an explicit one-dimensional integral; the piecewise powers above make it elementary after splitting at the angles where $L = L_{\text{Euc}}(q)$, at q_{dec} , and at $q = 2$.

The global contribution can also be read directly from the accessible volume:

$$\frac{dR_{\text{det}}}{d \ln L} = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{3} \left(\frac{L}{L_{\text{ref}}} \right)^{\alpha+1} \int_0^{q_{\text{nr}}} q \min \left[\left(\frac{L}{L_{\text{Euc}}(q)} \right)^{3/2}, 1 \right] dq. \quad (34)$$

For $L < L_{\text{dec}}$ no angle reaches the Euclidean boundary. The factor $L^{3/2}$ can be taken outside the integral, so the global logarithmic contribution rises as $L^{\alpha+5/2}$. For $L > L_{\text{nr}}$, all angles fill the volume and the global contribution falls as $L^{\alpha+1}$. Between these thresholds, different angular regions become fully visible in sequence.

The project's Euclidean angle $q_{\text{Euc}}(L)$ satisfies $L = L_{\text{Euc}}(q_{\text{Euc}})$. At this angle the flux horizon reaches D_{Euc} . Solving the physical threshold branches gives

$$q_{\text{Euc}}(L) = \begin{cases} 1 + (L/L_j)^{1/[2(p-1)]}, & L_{\text{dec}} \leq L < L_j, \\ 1 + (L/L_j)^{1/(2p)}, & L_j \leq L < L_{\text{nr}}. \end{cases} \quad (35)$$

All angles below $q_{\text{Euc}}(L)$ fill the volume; larger angles have smaller flux horizons. Thus the angular integral in Eq. (34), for $L_{\text{dec}} \leq L < L_{\text{nr}}$, is explicitly

$$\frac{q_{\text{Euc}}(L)^2}{2} + \begin{cases} \left(\frac{L}{L_j}\right)^{3/2} \left[\int_{q_{\text{Euc}}(L)}^2 q(q-1)^{-3(p-1)} dq + \int_2^{q_{\text{nr}}} q(q-1)^{-3p} dq \right], & L < L_j, \\ \left(\frac{L}{L_j}\right)^{3/2} \int_{q_{\text{Euc}}(L)}^{q_{\text{nr}}} q(q-1)^{-3p} dq, & L \geq L_j. \end{cases} \quad (36)$$

These integrals contain only powers of $q-1$: writing $q = (q-1) + 1$ evaluates them by exactly the method used below for the total rate. Equation (36) retains both the fully visible angular region and the more distant off-axis tail.

There is a useful further limit. In phase III, if $1 \ll q_{\text{Euc}}(L) \ll q_{\text{nr}}$, then $q-1 \simeq q$ and, for $p > 2/3$,

$$\int_{q_{\text{Euc}}(L)}^{q_{\text{nr}}} q^{1-3p} dq \simeq \frac{q_{\text{Euc}}(L)^{2-3p}}{3p-2}. \quad (37)$$

Since $q_{\text{Euc}}(L) \propto L^{1/(2p)}$, both terms in Eq. (36) scale as $L^{1/p}$. Consequently the global logarithmic luminosity contribution has the intermediate asymptote

$$\frac{dR_{\text{det}}}{d \ln L} \propto L^{\alpha+1+1/p}. \quad (38)$$

For $\alpha > -1 - 1/p$ it can still be rising as increasingly off-axis events become visible, even though every fixed-angle distribution already falls above its own threshold. Eventually $L > L_{\text{nr}}$ ends that angular gain and restores the declining power $L^{\alpha+1}$. This explains how the global distribution can favor much larger luminosities than L_{dec} .

Multiplying the luminosity integrand by L^n changes its convergence conditions to $-5/2 < \alpha + n < -1$, in addition to the original finite-rate condition. Thus the detected mean luminosity exists only for $-5/2 < \alpha < -2$. At $\alpha = -2$ its bright tail diverges logarithmically, even though the rate, median, and every fixed quantile are finite. Medians and enclosed fractions are consequently useful measures of a typical detection over much of the allowed slope interval.

5 Distance and viewing angle: two distinct dominance questions

5.1 The distance distribution

Equation (21) separates the distance and angle dependence:

$$\frac{d^2 R_{\text{det}}}{dD dq} = \frac{4\pi\theta_j^2 \mathcal{A}}{-\alpha-1} q \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} D^2 \left(\frac{D}{D_{\text{Euc}}} \right)^{2\alpha+2}. \quad (39)$$

The normalized distance density is therefore

$$\frac{1}{R_{\text{det}}} \frac{dR_{\text{det}}}{dD} = \frac{2\alpha+5}{D_{\text{Euc}}} \left(\frac{D}{D_{\text{Euc}}} \right)^{2\alpha+4}. \quad (40)$$

Integrating from 0 to d and then setting the result to one half gives

$$\boxed{\Pr(D < d \mid \text{detected}) = \left(\frac{d}{D_{\text{Euc}}} \right)^{2\alpha+5}, \quad D_{\text{med}} = D_{\text{Euc}} 2^{-1/(2\alpha+5)}}. \quad (41)$$

Per logarithmic distance interval, however,

$$\frac{1}{R_{\text{det}}} \frac{dR_{\text{det}}}{d \ln D} = (2\alpha+5) \left(\frac{D}{D_{\text{Euc}}} \right)^{2\alpha+5}. \quad (42)$$

This rises toward D_{Euc} throughout the convergent interval. The rise does not prove strong concentration at the boundary: the fraction beyond any specified distance d is only $1 - (d/D_{\text{Euc}})^{2\alpha+5}$. At $\alpha = -2$ the distribution per unit distance is uniform and $D_{\text{med}} = D_{\text{Euc}}/2$. As $\alpha \rightarrow -5/2^+$ the median tends to zero; as $\alpha \rightarrow -1^-$ it tends to $2^{-1/3}D_{\text{Euc}} \simeq 0.794D_{\text{Euc}}$.

The factorization in Eq. (39) also means that distance and viewing angle are independent *within the detected sample of this unbounded model*. Luminosity and distance are not independent. A change in limiting flux changes the selected luminosities and total rate, but leaves the normalized distance distribution unchanged. The same result survives a fixed cadence or random observing phase, as derived below. It fails in general when finite luminosity bounds, luminosity-dependent light-curve shapes, or distance-dependent selection introduce another scale.

5.2 The viewing-angle distribution

At each angle the remaining factor in Eq. (20) gives

$$\frac{1}{R_{\text{det}}} \frac{dR_{\text{det}}}{d \ln q} = \frac{q^2 [L_{\text{Euc}}(q)/L_{\text{ref}}]^{\alpha+1}}{\int_0^{q_{\text{nr}}} q' [L_{\text{Euc}}(q')/L_{\text{ref}}]^{\alpha+1} dq'}. \quad (43)$$

In the constant-peak-flux core it rises as q^2 . Outside the core the unnormalized logarithmic contribution is

$$q^2 \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} = \begin{cases} q^2 (L_{\text{dec}}/L_{\text{ref}})^{\alpha+1}, & q \leq q_{\text{dec}}, \\ q^2 (L_j/L_{\text{ref}})^{\alpha+1} (q-1)^{2(p-1)(\alpha+1)}, & q_{\text{dec}} < q < 2, \\ q^2 (L_j/L_{\text{ref}})^{\alpha+1} (q-1)^{2p(\alpha+1)}, & 2 \leq q \leq q_{\text{nr}}. \end{cases} \quad (44)$$

There is a competition between the growing number of orientations (q^2 per logarithmic interval) and the larger luminosities needed at larger angles. Because $\alpha + 1 < 0$, increasing the threshold reduces the number of qualifying events.

For example, the exact logarithmic slope in phase III is

$$\frac{d \ln [q^2 (q-1)^{2p(\alpha+1)}]}{d \ln q} = 2 + 2p(\alpha+1) \frac{q}{q-1}. \quad (45)$$

For $q \gg 1$, the contribution behaves as $q^{2+2p(\alpha+1)}$. It declines in that limit only for $\alpha < -1 - 1/p$. This is a condition on the off-axis tail, not on the luminosity integral: the total rate remains convergent throughout $-5/2 < \alpha < -1$ because q_{nr} is finite. Near $\alpha = -1$ the luminosity penalty becomes weak and the approximate orientation measure itself controls the angular distribution. Since $\theta_j q_{\text{nr}} = \sqrt{2}$ rad, that extreme regime also presses against the physical accuracy of the project's small-angle geometry.

Concentration near $q \simeq 1$ and concentration near D_{Euc} are separate claims. A steep luminosity function can strongly favor small viewing angles while placing most detections nearby. A shallow one can favor the outer volume while giving a substantial off-axis contribution.

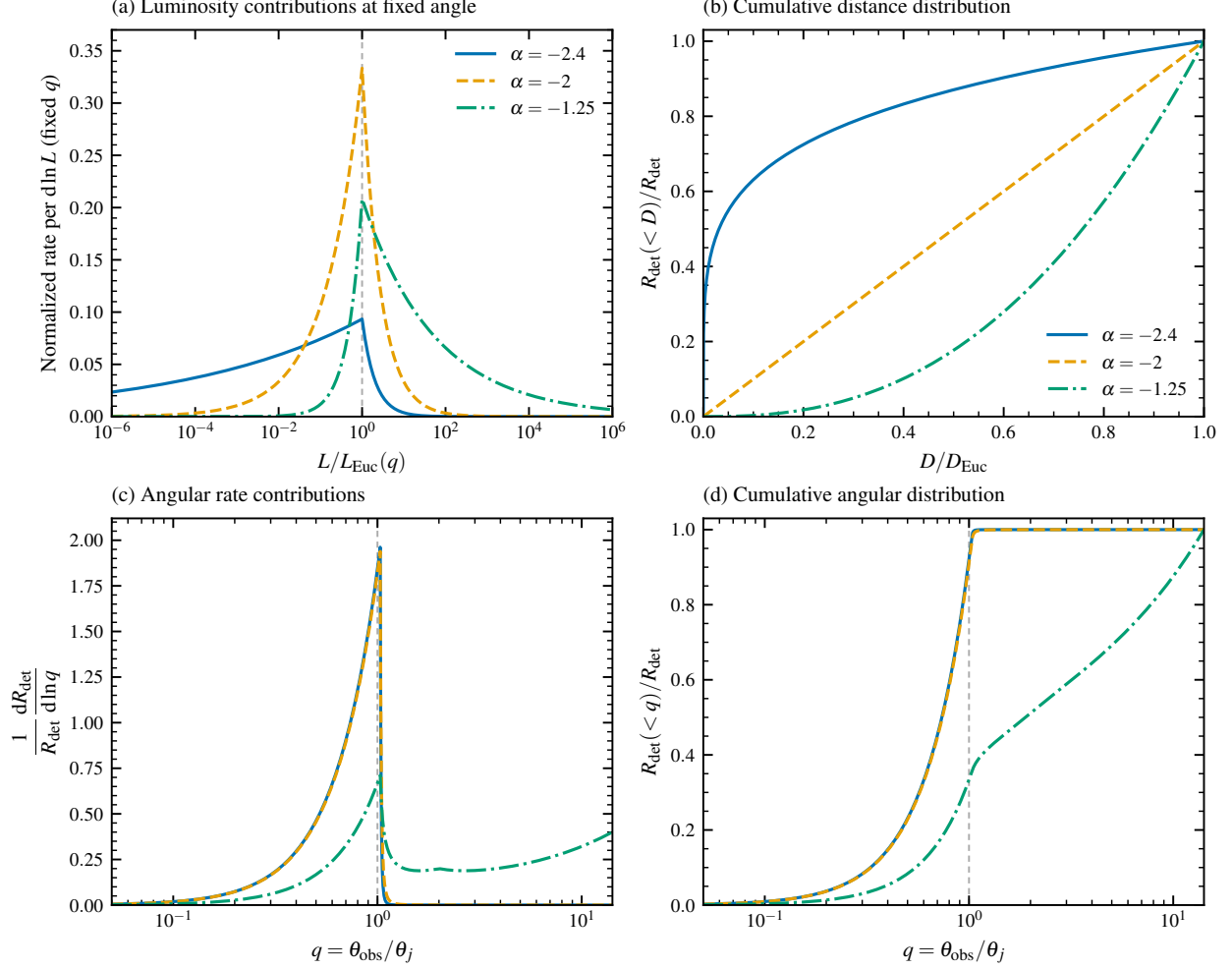


Figure 1: General detected-population behavior for three LF slopes. The luminosity panel is conditional on a fixed angle and uses its physical threshold $L_{\text{Euc}}(q)$. Its plotted range does not include all of the long tails; normalization includes the full range $0 < L < \infty$. The distance distribution is independent of the light-curve shape. The angular panels use continuous detection with $p = 2.5$, $\theta_j = 0.1$, and $\Gamma_0 = 10^{2.5}$. Areas under logarithmic densities are taken with respect to their logarithmic horizontal coordinates.

6 Explicit angular integration and the corner approximation

Using the three branches of Eq. (13), the remaining integral is

$$\begin{aligned} \int_0^{q_{\text{nr}}} q \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} dq &= \frac{q_{\text{dec}}^2}{2} \left(\frac{L_{\text{dec}}}{L_{\text{ref}}} \right)^{\alpha+1} \\ &+ \left(\frac{L_{\text{j}}}{L_{\text{ref}}} \right)^{\alpha+1} \int_{q_{\text{dec}}}^2 q(q-1)^{2(p-1)(\alpha+1)} dq \\ &+ \left(\frac{L_{\text{j}}}{L_{\text{ref}}} \right)^{\alpha+1} \int_2^{q_{\text{nr}}} q(q-1)^{2p(\alpha+1)} dq. \end{aligned} \quad (46)$$

To integrate, write $q = (q-1) + 1$. Each integrand is then the sum of two powers of $q-1$. Carrying out all three pieces gives the full rate:

$$\begin{aligned} R_{\text{det}} &= \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{(-\alpha-1)(2\alpha+5)} \left\{ \frac{q_{\text{dec}}^2}{2} \left(\frac{L_{\text{dec}}}{L_{\text{ref}}} \right)^{\alpha+1} \right. \\ &+ \left(\frac{L_{\text{j}}}{L_{\text{ref}}} \right)^{\alpha+1} \left[\frac{1 - (q_{\text{dec}} - 1)^{2+2(p-1)(\alpha+1)}}{2 + 2(p-1)(\alpha+1)} \right. \\ &+ \frac{1 - (q_{\text{dec}} - 1)^{1+2(p-1)(\alpha+1)}}{1 + 2(p-1)(\alpha+1)} \\ &+ \frac{(q_{\text{nr}} - 1)^{2+2p(\alpha+1)} - 1}{2 + 2p(\alpha+1)} \\ &\left. \left. + \frac{(q_{\text{nr}} - 1)^{1+2p(\alpha+1)} - 1}{1 + 2p(\alpha+1)} \right] \right\}. \end{aligned} \quad (47)$$

This expression is exact within the project's light curve and angular geometry. It includes all luminosities and distances, as well as the continuous transition across the jet break at $q = 2$.

A denominator can vanish without the integral diverging. For the phase-II terms this occurs at $\alpha = -1 - 1/(p-1)$ or $\alpha = -1 - 1/[2(p-1)]$; replace the corresponding quotient by $-\ln(q_{\text{dec}} - 1)$. For the phase-III terms it occurs at $\alpha = -1 - 1/p$ or $\alpha = -1 - 1/(2p)$; replace the corresponding quotient by $\ln(q_{\text{nr}} - 1)$. These are ordinary logarithmic integrals of $(q-1)^{-1}$, with finite positive limits. They are unrelated to the population-divergence boundaries $\alpha = -5/2$ and $\alpha = -1$.

6.1 What is lost by retaining only the $q \simeq 1$ region?

Keeping only the constant-peak-flux core means retaining just the first term in braces in Eq. (47):

$$R_{\text{det}}(q \leq q_{\text{dec}}) = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{(-\alpha-1)(2\alpha+5)} \frac{q_{\text{dec}}^2}{2} \left(\frac{L_{\text{dec}}}{L_{\text{ref}}} \right)^{\alpha+1}. \quad (48)$$

Dividing this by the full rate gives exactly the detected fraction in that core. Its complement is exactly the fractional rate omitted by this approximation. This provides a direct accuracy measure rather than an assumption of dominance. The geometrical on-axis region ends at $q = 1$; the constant-peak-flux core ends at the slightly larger q_{dec} .

The thesis's most-contributing rectangle is a different approximation: it changes its angle with luminosity. For $L < L_{\text{dec}}$ it retains the core out to its flux horizon; for $L > L_{\text{dec}}$ it retains angles whose entire Euclidean column is visible. Thus it keeps the core contribution and, at every outer angle, only the luminosities above $L_{\text{Euc}}(q)$. Using Eq. (30) gives

$$\frac{R_{\text{det}}^{\text{rectangle}}}{R_{\text{det}}} = 1 - \frac{2(-\alpha-1)}{3} \left[1 - \frac{R_{\text{det}}(q \leq q_{\text{dec}})}{R_{\text{det}}} \right]. \quad (49)$$

This explicitly separates the omitted outer-angle population from the fraction of that population whose flux horizon is inside D_{Euc} . A good rectangle estimate does not by itself imply that most detections have $q \simeq 1$.

7 How repeated detections change the threshold

7.1 The thesis cadence approximation

Let t_+ be the time at which the declining light curve falls below the survey limit. For $i \geq 2$, the thesis step approximation requires both a detectable peak and $t_+ \geq it_{\text{cad}}$, after approximating the visible duration by t_+ . At D_{Euc} , the latter condition requires

$$L \geq L_i \equiv \frac{L_{\text{dec}}}{\tilde{F}_\nu(it_{\text{cad}})}. \quad (50)$$

Writing it out in physical times,

$$L_i = \begin{cases} L_{\text{dec}}, & it_{\text{cad}} \leq t_{\text{dec}}, \\ L_{\text{dec}}(it_{\text{cad}}/t_{\text{dec}})^{3(p-1)/4}, & t_{\text{dec}} < it_{\text{cad}} < t_j, \\ L_j(it_{\text{cad}}/t_j)^p, & t_j \leq it_{\text{cad}} < t_{\text{nr}}, \\ L_{\text{nr}}(it_{\text{cad}}/t_{\text{nr}})^{(15p-21)/10}, & it_{\text{cad}} \geq t_{\text{nr}}. \end{cases} \quad (51)$$

Peak detection requires $L \geq L_{\text{Euc}}(q)$; repeated detection in this approximation requires $L \geq L_i$. The actual threshold is therefore

$$\boxed{L \geq \max[L_{\text{Euc}}(q), L_i] \text{ at } D_{\text{Euc}}.} \quad (52)$$

At a smaller distance, multiply that threshold by $(D/D_{\text{Euc}})^2$. The derivation of the previous sections consequently applies with the explicit replacement $L_{\text{Euc}}(q) \rightarrow \max[L_{\text{Euc}}(q), L_i]$:

$$R_{\geq i}^{\text{thesis}} = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{(-\alpha-1)(2\alpha+5)} \int_0^{q_{\text{nr}}} q \left(\frac{\max[L_{\text{Euc}}(q), L_i]}{L_{\text{ref}}} \right)^{\alpha+1} dq. \quad (53)$$

The convergence interval and the exact F_{lim} and D_{Euc} scalings are unchanged. The luminosity and angle distributions change because the physical threshold has changed.

The angle q_i at which the two constraints agree satisfies $t_p(q_i) = it_{\text{cad}}$, hence

$$q_i = \begin{cases} 1 + (it_{\text{cad}}/t_j)^{3/8}, & t_{\text{dec}} < it_{\text{cad}} < t_j, \\ 1 + (it_{\text{cad}}/t_j)^{1/2}, & t_j \leq it_{\text{cad}} < t_{\text{nr}}. \end{cases} \quad (54)$$

For $q < q_i$ the common threshold is L_i ; for $q > q_i$ it is $L_{\text{Euc}}(q)$. The candidate most-contributing point therefore moves from $(q_{\text{dec}}, D_{\text{Euc}})$ at luminosity L_{dec} to (q_i, D_{Euc}) at luminosity L_i . In particular, q_i need not be close to one.

7.2 The cadence integral is still analytic

For $q_{\text{dec}} < q_i < 2$, Eq. (47) applies after replacing q_{dec} by q_i in the angular limits and replacing L_{dec} by L_i in the core term; L_j is unchanged. This directly performs all the integrals in Eq. (53). For $2 \leq q_i < q_{\text{nr}}$, the phase-II integral is absent and the result is

$$R_{\geq i}^{\text{thesis}} = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{(-\alpha-1)(2\alpha+5)} \left\{ \frac{q_i^2}{2} \left(\frac{L_i}{L_{\text{ref}}} \right)^{\alpha+1} + \left(\frac{L_j}{L_{\text{ref}}} \right)^{\alpha+1} \left[\frac{(q_{\text{nr}}-1)^{2+2p(\alpha+1)} - (q_i-1)^{2+2p(\alpha+1)}}{2+2p(\alpha+1)} + \frac{(q_{\text{nr}}-1)^{1+2p(\alpha+1)} - (q_i-1)^{1+2p(\alpha+1)}}{1+2p(\alpha+1)} \right] \right\}. \quad (55)$$

A zero denominator again gives the logarithm $\ln[(q_{\text{nr}} - 1)/(q_i - 1)]$ for the relevant quotient. For $it_{\text{cad}} \leq t_{\text{dec}}$ the continuous result is recovered. For $it_{\text{cad}} \geq t_{\text{nr}}$, all angles are time limited and the integral in Eq. (53) is simply $(q_{\text{nr}}^2/2)(L_i/L_{\text{ref}})^{\alpha+1}$.

There is no single cadence power law valid across all these cases. In phase III, if $1 \ll q_i \ll q_{\text{nr}}$ and $-p(\alpha + 1) > 1$, the outer-angle integral is dominated by its lower limit and the rate scales approximately as

$$R_{\geq i}^{\text{thesis}} \propto (it_{\text{cad}})^{1+p(\alpha+1)}. \quad (56)$$

The extra power of one comes from $q_i^2 \propto it_{\text{cad}}$; the other factor comes from $L_i^{\alpha+1} \propto (it_{\text{cad}})^{p(\alpha+1)}$. The conditions matter: near the jet break, near the outer angular boundary, or for a shallow luminosity function, the explicit expression must be used. At fixed shape, increasing the required duration cannot increase the exact rate, because it only raises the threshold in Eq. (53).

7.3 Retaining the peak time and the random observing phase

The physical visible duration is $\Delta t_{\text{det}} = t_+ - t_p(q)$ when positive. In a periodic sequence of instantaneous visits, the next visit occurs at a uniformly random time between $t_p(q)$ and $t_p(q) + t_{\text{cad}}$. The time of the i th visit is therefore uniform between $t_p(q) + (i-1)t_{\text{cad}}$ and $t_p(q) + it_{\text{cad}}$. The project probability ramp follows:

$$P_{\geq i} = \begin{cases} 0, & \Delta t_{\text{det}} < (i-1)t_{\text{cad}}, \\ \Delta t_{\text{det}}/t_{\text{cad}} - (i-1), & (i-1)t_{\text{cad}} \leq \Delta t_{\text{det}} < it_{\text{cad}}, \\ 1, & \Delta t_{\text{det}} \geq it_{\text{cad}}. \end{cases} \quad (57)$$

At a specified time t of that i th visit, the required luminosity at D_{Euc} is $L_{\text{dec}}/\tilde{F}_\nu(t)$. Integrating the luminosity and distance exactly as before, then averaging over the visit time, gives

$$R_{\geq i}^{\text{random phase}} = \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{(-\alpha-1)(2\alpha+5)} \int_0^{q_{\text{nr}}} q dq \frac{1}{t_{\text{cad}}} \times \int_{t_p(q)+(i-1)t_{\text{cad}}}^{t_p(q)+it_{\text{cad}}} \left(\frac{L_{\text{dec}}}{L_{\text{ref}}\tilde{F}_\nu(t)} \right)^{\alpha+1} dt. \quad (58)$$

The time integral is a power or logarithm on each temporal branch; its explicit primitives are given in Appendix A. Splitting at t_j and t_{nr} handles visits crossing a transition. The final angular integral can then be evaluated by one-dimensional quadrature. No additional luminosity or distance quadrature is needed.

Since the light curve decreases, evaluating the inner integrand at its latest endpoint gives a lower bound on its average; evaluating it at its earliest endpoint gives an upper bound. These are the two peak-retained step bounds on the random-phase rate. The thesis step instead uses the threshold time $\max[t_p(q), it_{\text{cad}}]$. It is a different approximation, so there is no general ordering between it and the random-phase result. For $i = 1$, peak detectability alone is not enough for a periodic visit to catch the event. As $t_{\text{cad}} \rightarrow 0$ at fixed i , Eq. (58) recovers continuous detection.

Crucially, the visit times depend on angle and cadence but not on D or L . The distance factor in Eq. (21) is therefore unchanged: all these prescriptions share Eq. (41). The same is true for any selection depending on flux, viewing angle, and timing, provided it introduces no separate luminosity or distance scale and has nonzero finite selected weight. Two such surveys can have different rates and luminosity distributions while predicting the same normalized distance distribution.

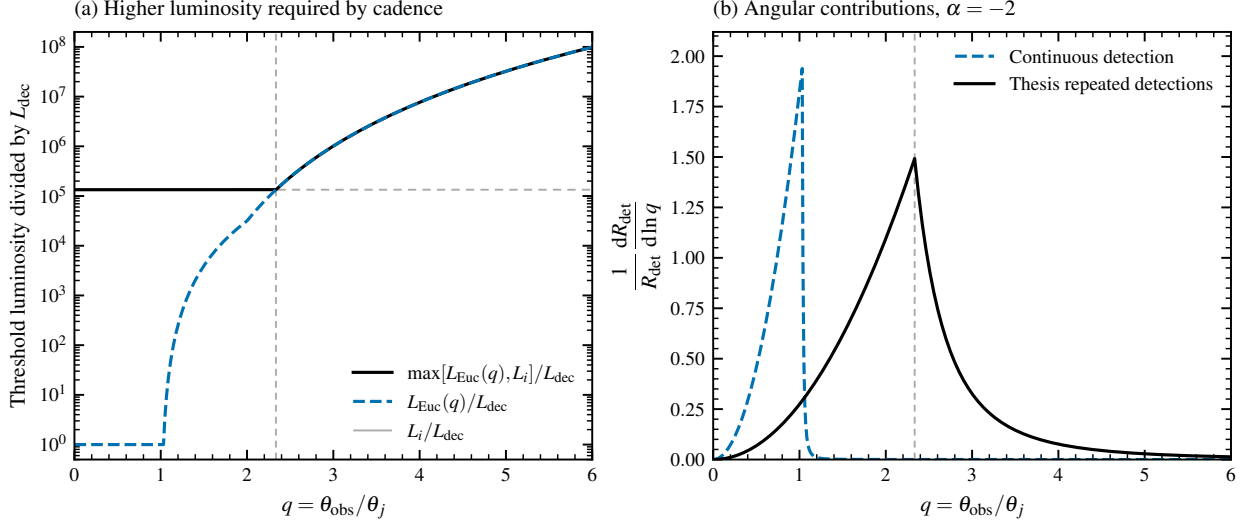


Figure 2: How repeated detections alter the threshold and viewing angles. The thesis step replaces $L_{\text{Euc}}(q)$ by $\max[L_{\text{Euc}}(q), L_i]$. Using the shape in Figure 1, $t_{\text{dec}} = 19.4$ s, $\alpha = -2$, $i = 2$, and a two-day cadence, the angular contribution rises to the shifted corner at $q_i = 2.335$. These curves illustrate the analytic step construction, not a precision selection model for a particular survey.

8 Relation to the original bounded luminosity function

8.1 Match the differential normalization, not the total rate

For the same intensity restricted to $L_{\min} \leq L \leq L_{\max}$, the finite intrinsic rate density is

$$\mathcal{R}_{[L_{\min}, L_{\max}]} = \frac{\mathcal{A}}{\alpha + 1} \left[\left(\frac{L_{\max}}{L_{\text{ref}}} \right)^{\alpha+1} - \left(\frac{L_{\min}}{L_{\text{ref}}} \right)^{\alpha+1} \right]. \quad (59)$$

The normalized distribution used by the previous model is

$$\phi(L) = \frac{(\alpha + 1)L^\alpha}{L_{\max}^{\alpha+1} - L_{\min}^{\alpha+1}}, \quad L_{\min} \leq L \leq L_{\max}. \quad (60)$$

Equations (59) and (60) satisfy $\mathcal{R}_{[L_{\min}, L_{\max}]} \phi(L) = d\mathcal{R}/dL$. The previous total rate inside the Euclidean sphere is then $R_{\text{int}} = (4\pi/3)D_{\text{Euc}}^3 \mathcal{R}_{[L_{\min}, L_{\max}]}$. At $\alpha = -1$, the intrinsic normalization is instead $\mathcal{A} \ln(L_{\max}/L_{\min})$ and $\phi(L) = 1/[L \ln(L_{\max}/L_{\min})]$.

The correct infinite-range limit holds $\mathcal{A}$ fixed while moving the bounds outward. The finite total intrinsic rate must then change. Holding R_{int} fixed while lowering $L_{\min}$ would reduce the number of events at every fixed detectable luminosity for $\alpha < -1$. That procedure tests a changing population, not convergence to the simplified intensity.

8.2 How rapidly do finite-cutoff results approach this model?

Suppose first that $L_{\min} < L_{\text{Euc}}(q) < L_{\max}$ at a given angle. Integrating the two sides of Eq. (16) with finite limits gives

$$\begin{aligned} \frac{dR_{\text{det}}^{\text{bounded}}}{dq} &= \frac{4\pi\theta_j^2 D_{\text{Euc}}^3 \mathcal{A}}{3} q \left(\frac{L_{\text{Euc}}(q)}{L_{\text{ref}}} \right)^{\alpha+1} \\ &\times \left[\frac{1 - (L_{\min}/L_{\text{Euc}}(q))^{\alpha+5/2}}{\alpha + 5/2} + \frac{1 - (L_{\max}/L_{\text{Euc}}(q))^{\alpha+1}}{-\alpha - 1} \right]. \end{aligned} \quad (61)$$

If both bounds lie below $L_{\text{Euc}}(q)$, only the unsaturated volume branch is present. If both lie above it, only the fully visible branch is present. In either case the original volume integral, Eq. (15), gives the appropriate finite limits without extrapolating Eq. (61).

For a threshold between the bounds, the fractional contributions excluded by the lower and upper cutoffs are respectively

$$\frac{dR_{\text{det}}(L < L_{\min})/dq}{dR_{\text{det}}/dq} = \frac{2(-\alpha - 1)}{3} \left(\frac{L_{\min}}{L_{\text{Euc}}(q)} \right)^{\alpha+5/2}, \quad (62)$$

$$\frac{dR_{\text{det}}(L > L_{\max})/dq}{dR_{\text{det}}/dq} = \frac{2(\alpha + 5/2)}{3} \left(\frac{L_{\max}}{L_{\text{Euc}}(q)} \right)^{\alpha+1}. \quad (63)$$

These follow equally from the conditional CDF. Their detected angular-weighted average is the fractional error in the total rate. For cadence, use $\max[L_{\text{Euc}}(q), L_i]$; for random phase, also average over the selected visit times.

The tail powers approach zero at the convergence boundaries, so even very remote cutoffs can then matter. At $\alpha = -5/2$, the faint integral is proportional to $\ln[L_{\text{Euc}}(q)/L_{\min}]$; at $\alpha = -1$, the bright integral is proportional to $\ln[L_{\max}/L_{\text{Euc}}(q)]$. Outside the convergent interval the divergence is a power of the corresponding cutoff. There is no cutoff-independent detected-rate limit there.

8.3 What changes in the dependence on survey depth?

A bounded population has two limiting behaviors at fixed cadence and shape. If the survey is so shallow that all its luminosities have horizons inside D_{Euc} , the volume factor gives $R_{\text{det}}^{\text{bounded}} \propto F_{\text{lim}}^{-3/2}$. If it is so deep that every luminosity at every included angle fills the Euclidean volume, the rate is independent of F_{lim} . In between, some luminosity-angle pairs are unsaturated and others are already visible throughout the volume.

Differentiating Eq. (15) makes this precise. Only its unsaturated branch depends on F_{lim} , with logarithmic derivative $-3/2$. The moving-boundary terms cancel because both volume branches agree at $L = L_{\text{Euc}}(q)$. Thus, for a fixed bounded population under continuous detection,

$$\frac{d \ln R_{\text{det}}^{\text{bounded}}}{d \ln F_{\text{lim}}} = -\frac{3}{2} \frac{R_{\text{det}}^{\text{bounded}}(L < L_{\text{Euc}}(q))}{R_{\text{det}}^{\text{bounded}}}. \quad (64)$$

The numerator counts the part of the selected rate whose flux horizon is inside the volume, integrating over both luminosity and angle. In the unbounded power law its fraction is the same at every angle, $2(-\alpha - 1)/3$, from Eq. (29). Consequently Eq. (64) reduces exactly to the slope $\alpha + 1$. For the thesis cadence step the same derivation uses the unsaturated condition $L < \max[L_{\text{Euc}}(q), L_i]$.

This explains the simplification physically. A bounded LF moves through shallow, intermediate, and saturated survey regimes. An unbounded power law always supplies fainter and brighter events around the current threshold, so the balance between unsaturated and saturated contributions stays fixed. Its single exact flux power law describes the intermediate behavior when both real luminosity bounds are sufficiently remote; it cannot reproduce the two eventual cutoff-controlled limits.

9 A few illustrations and the limits of the approximation

Table 1 uses only the analytic expressions above. It does not require an afterglow parameter set or a calibrated normalization. The contrast between a mode and an enclosed fraction is already visible.

Table 1: Dimensionless examples from Eqs. (29), (31), and (41). Luminosity statements are conditional on a fixed viewing angle; the distance median is global.

α	$L_{\text{med} q}/L_{\text{Euc}}(q)$	$\Pr(L < L_{\text{Euc}}(q) \mid q, \text{detected})$	$D_{\text{med}}/D_{\text{Euc}}$
-2.4	0.00195	0.933	0.03125
-2	0.5625	0.667	0.500
-1.25	7.716	0.167	0.758

For the steeper slope the threshold remains the conditional logarithmic maximum, yet most detections are much fainter and closer than that threshold might suggest. For the shallower slope most events at each angle are above the local threshold. Their typical luminosity grows further when more off-axis angles, with larger thresholds, contribute.

Figure 1 adds the project’s illustrative shape $p = 2.5$, $\theta_j = 0.1$, $\Gamma_0 = 10^{2.5}$, and $t_{\text{dec}} = 19.4$ s. At $\alpha = -2$, about 96.9% of continuously detected events lie within $q_{\text{dec}} = 1.0316$. Thus this case supports the angular-core approximation, while its distance median is still only $D_{\text{Euc}}/2$. With two required detections separated by two days, the same shape gives $q_i = 2.335$ and $L_i/L_{\text{dec}} \simeq 1.34 \times 10^5$. Figure 2 illustrates why the original $q \simeq 1$ threshold is then no longer the relevant repeated-detection threshold.

Retaining the peak and averaging phase need not be a small correction to the thesis step. For this same two-day example at $\alpha = -2$, the rate is about 14% higher for two required visits, but about 48% lower for ten required visits. The change of sign illustrates why a universal correction factor is unsuitable. These are examples of the distinction between timing prescriptions, not fitted observational results.

The analytic calculations were checked against independent quadrature, including the logarithmic cases, CDF normalization, recovered medians, and reference-luminosity invariance. Smooth controlled checks agree to better than the target relative accuracy 10^{-7} . Finite-cutoff calculations matched to the unchanged bounded-LF engine agree within 0.042% in rate and 0.025% in distance median for the stored reference cases, comfortably inside the one-percent criterion. Those comparisons keep the differential normalization fixed as the bounds move. The numerical tests, resolution studies, and the earlier ZTF benchmark remain in the accompanying analysis files; their detailed tables are not needed for this derivation.

10 What can be taken into the paper?

The two-parameter intensity gives a closed detected-rate calculation because the detectable volume rises as $L^{3/2}$ and then saturates. Those two behaviors yield the convergence interval, the exact flux law, and the two sides of the detected luminosity distribution directly.

The threshold $L_{\text{dec}} = 4\pi D_{\text{Euc}}^2 F_{\text{lim}}$ has a precise role: it is the conditional logarithmic maximum within the full-time angular core. Calling it the characteristic detected luminosity additionally requires checking the breadth of the distribution and the contribution of other angles. Cadence changes that threshold to L_i within q_i , and the random-phase construction gives a controlled way to retain the peak time.

The distance prediction is especially restrictive. At fixed α and D_{Euc} , changes in survey depth and homogeneous timing selection do not change the normalized distance distribution. The bounded model can depart from this prediction when a luminosity cutoff becomes relevant. This provides a general comparison between the models, separate from any particular normalization or survey benchmark.

The simplification is therefore useful as a controlled description of the detected population when omitted luminosity tails are small. It is not a complete intrinsic population model. The explicit tail fractions identify when physical cutoffs can be ignored and when they must be restored.

A Explicit time integrals for random-phase detection

To evaluate Eq. (58), split its physical time interval at t_j and t_{nr} . If the two endpoints t_1, t_2 lie within phase II,

$$\int_{t_1}^{t_2} \left(\frac{L_{\text{dec}}}{L_{\text{ref}} \tilde{F}_\nu(t)} \right)^{\alpha+1} dt = t_{\text{dec}} \left(\frac{L_{\text{dec}}}{L_{\text{ref}}} \right)^{\alpha+1} \frac{(t_2/t_{\text{dec}})^{1+3(p-1)(\alpha+1)/4} - (t_1/t_{\text{dec}})^{1+3(p-1)(\alpha+1)/4}}{1 + 3(p-1)(\alpha+1)/4}. \quad (65)$$

Within phase III the same integral is

$$\int_{t_1}^{t_2} \left(\frac{L_{\text{dec}}}{L_{\text{ref}} \tilde{F}_\nu(t)} \right)^{\alpha+1} dt = t_j \left(\frac{L_j}{L_{\text{ref}}} \right)^{\alpha+1} \frac{(t_2/t_j)^{1+p(\alpha+1)} - (t_1/t_j)^{1+p(\alpha+1)}}{1 + p(\alpha+1)}. \quad (66)$$

Within phase IV it is

$$\int_{t_1}^{t_2} \left(\frac{L_{\text{dec}}}{L_{\text{ref}} \tilde{F}_\nu(t)} \right)^{\alpha+1} dt = t_{\text{nr}} \left(\frac{L_{\text{nr}}}{L_{\text{ref}}} \right)^{\alpha+1} \frac{(t_2/t_{\text{nr}})^{1+(15p-21)(\alpha+1)/10} - (t_1/t_{\text{nr}})^{1+(15p-21)(\alpha+1)/10}}{1 + (15p-21)(\alpha+1)/10}. \quad (67)$$

When the denominator of one expression vanishes, replace that expression by, respectively,

$$t_{\text{dec}}(L_{\text{dec}}/L_{\text{ref}})^{\alpha+1} \ln(t_2/t_1), \quad t_j(L_j/L_{\text{ref}})^{\alpha+1} \ln(t_2/t_1), \quad t_{\text{nr}}(L_{\text{nr}}/L_{\text{ref}})^{\alpha+1} \ln(t_2/t_1). \quad (68)$$

These are finite-interval logarithmic limits. The Newtonian branch changes the timing weight when a required visit crosses t_{nr} ; it does not change the luminosity–distance convergence interval.

B Source conventions and reproducibility

The thesis and the previous LF derivation set the notation. The following source discrepancies are resolved using the physical light curve:

1. The jet-break threshold follows from the decreasing peak flux: $\tilde{F}_{\text{lim},j} = (\Gamma_0 \theta_j)^{-2(p-1)}$. This corrects the sign in thesis Eq. 45.
2. The phase-III light curve can also be written as $\tilde{F}_\nu(t) = (t_j/t_{\text{dec}})^{(p+3)/4} (t/t_{\text{dec}})^{-p}$. This fixes the jet-time factor in thesis Eq. 69.
3. The single-luminosity phase-III cadence factor contains $(it_{\text{cad}}/t_j)^{-3p/2} [1 + (it_{\text{cad}}/t_j)^{1/2}]^2$. Its large-duration power is $1-3p/2$, not $-3(p-2)/2$. Equation (56) is the separate LF-integrated asymptote.
4. The probability of one visit is $\min(\Delta t_{\text{det}}/t_{\text{cad}}, 1)$, following the thesis ramp. A printed maximum that exceeds unity cannot be used as a probability.
5. The reference engine continues phase III in its timing inversion; the thesis includes the Newtonian branch. Equation (7) uses the thesis branch. Finite-cutoff engine checks compare matching timing conventions, and the separate late-time calculations retain this distinction.

For comparisons to the application’s one-day luminosity, use

$$\nu L_\nu(1 \text{ day}) = \nu \tilde{F}_\nu(1 \text{ day}) L_\nu(t_{\text{dec}}). \quad (69)$$

The broken light curve is essential if the jet break precedes one day. This linear transformation preserves α , but the luminosity limits and normalization must refer to consistent units and epochs.

Calculations, tests, and rebuild instructions are in `analysis/simplified_luminosity_function/`. The app and engine are unchanged.

References

- [1] Bachelor's project, *Detection Rates of Gamma-Ray Burst Afterglows in Optical Surveys – An Analytic Approach*. Local source in `sources/`; especially Eqs. 21, 32–40, 45–53, 60, 63–77 and Figure 15.
- [2] *Detection Rates of GRB Afterglows with a Luminosity Distribution*. Project derivation, September 2026, `docs/luminosity_function_derivation.tex`.